

## Lesson 6

Equipment: Raspberry Pi Python code editor, provided sheets, pens/pencils, post-it notes

### Lesson Aims

- I can make predictions about the execution of code written by my peers
- I can design and plan a program in a team
- I can work in a team to create a program

### Curriculum Integration

- TCH 2-13a: pupils first write their code on paper and should reflect on changes necessary to get their artwork to look as expected. This requires understanding of the process they have defined in code. Additionally, planning their program involves devising an appropriate program structure.
- TCH 2-14a: the planning process requires an understanding of basic Python turtle commands to be able to “translate” ideas into a program.
- TCH 2-15a: group project challenges pupils to design their own piece of artwork using Python’s turtle module. Pupils also take part in a review process where they evaluate the code produced by other teams and have their own code evaluated.

### Plan

#### Starter

- Prompt: To think about: why do we plan code before we write it?

#### Unplugged (5-10 minutes)

- Pupils will be given a short program. Three possible descriptions of this program can be seen on slide 4.
- Pupils should be given a few minutes to discuss as a team which of the three descriptions they think matches the given program.

#### Taught section (5-10 minutes)

- This will be a shorter section than in previous lessons. This is flexible and dependent on your own perceptions of pupil understanding of various turtle functions. This should be used to demonstrate the use of turtle functions such as `turtle.color()`, `turtle.begin_fill()/turtle.end_fill()` and `turtle.pensize()`.

#### Coding activity (15 minutes planning, 30 minutes coding)

- Pupils should be given 15 minutes to plan their programs as a group. This should involve drawing a basic picture of the program they are aiming to create. Additionally, pupils should work in teams to write on paper the code their team is

planning to write. Teams will be provided with cheatsheets with descriptions of turtle functions and they should be encouraged to use these.

- It is ok for teams who are struggling with this task to write more generic descriptions for their program plan. For example, writing a list of commands they think will be useful for a particular purpose is also acceptable.
- Pupils then have 30 minutes to work on their code in their teams. Pupils should be encouraged to make notes on their sheets of where they had to make changes to their code from their plans. This may be challenging to implement in practise. Potentially, allowing time at the end of the coding process for pupils to compare their final code to their plan could be an effective way to implement this.

## **Closing**

- Pupils will review another group's program.
- Pupils should first make a prediction of what they think the program of another team will draw before actually running the program.
- Pupils will then write 2 short comments for the other team. One thing that they liked about the program and one thing that could be improved.
- [Optional] If you feel your class would respond well to this, a mini competition could be held. Teams can view the programs created by all other groups and vote for their favourite. The team with the most votes wins.

## **Spatial skills**

- There is a spatial visualisation process involved in pupils matching a program to the correct description. Pupils should be encouraged to visualise the execution of the program and it should be ensured they are not using their laptops during this task.
- Additionally, pupils should follow a VPC process in the development of their projects as they first write their code on paper then compare this to their finished code.

## **General teaching advice**

- Pupils should be strongly encouraged to use their cheatsheets. Being able to recognise how a particular command could be utilised in their own program is a valuable skill and could be argued to contribute to their spatial skills.
- Pupils will likely be enthusiastic about their projects and it may be difficult to shift their focus onto the latter tasks involving reflecting on their own and others' code. Encourage their participation in the review process as this part of the exercise is aimed at training their spatial skills.